MIL-V-23151B(SHIPS)
10 April 1967
SUPERSEDING
MIL-V-23151A(SHIPS)
2 April 1962
(See 6.6)

MILITARY SPECIFICATION

VOLTMETER, EXPANDED SCALE SWITCHBOARD TYPE

(NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope. - This specification covers high accuracy, shockproof switchboard mounted, expanded scale voltmeters.

MR	49	W
Component (1.2.1.1)	Style (1.2.1.2)	Color scheme (1.2.1.3)

1.2 Classification. -

1.2.1 Type designation. - The type designation of voltmeters shall be in the following form, and as specified (see table II and 6.2):

115	AF	H
Center-scale value (1.2.1.4)	Kind of current (1.2.1.5)	Shockproof feature (1.2.1.6)

Center-scale value	Scale range
115	110-120
120	115-125
208	198-218
220	210-230
440	420-460
450	430-470

1.2.1.1 Component. - Voltmeters covered by this specification shall be designated by the two letter symbol "MR".

1.2.1.2 Style. - The style shall be designated by a two digit number which signifies the following characteristics:

1.2.1.2.1 First digit. - The first digit shall designate the nominal size and shape of flange as follows:

3 - 3-1/2-inch round flange, sealed.
4 - 4-1/2-inch rectangular flange.

1.2.1.2.2 Second digit. - The second digit shall designate the type of mounting and the nominal size of length in angular degrees, as follows:

9 flush mounting, 250 degree nominal scale.

FSC 6625

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1.2.1.3 Color scheme. - The color scheme of the dial background, markings, and pointer shall be designated by a single letter as follows:

W - White dial background, black markings and pointer.

1.2.1.4 Center-scale value. - The center-scale value shall be designated by three digits for the scale ranges covered by this specification as follows;

1.2.1.5 Kind of current. - Two letters shall designate the kind of current for which the meter is designed as follows:

AF - Alternating current (ac) 400 cycle per second (cps) nominal frequency.

1.2.1.6 Shockproof feature. - A single letter shall designate the shockproof feature as follows:

H - High-impact shock resistant.

1.2.1.7 Example of type designation. - The type designation MR39W115AFH signifies the following:

(a) MR - voltmeter electrical indicating.
(b) 39 - 3-1/2 inch round flange sealed flush mounting 250 degree nominal scale.
(c) W - White dial background with black markings and black pointer.
(d) 115 - Center-scale value 115 volts, 110 to 120 volt range.
(e) AF - Ac 400 cps.
(f) H - High-impact shockproof.

2. APPLICABLE DOCUMENTS

2.1. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic-Parts, Thermosetting.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-M-10304 - Meters, Electrical Indicating Panel Type, Ruggedized, General Specification for.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)

C39.1-1964 - Electrical Indicating Instruments, Requirements for.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Qualification. - Voltmeters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Definitions. - Definitions of terms used in the specification shall be in accordance with USAS C39.1, except as specified herein.

3.2.1 Accuracy. - The accuracy is a number which defines the limit of error, expressed as a percentage of center-scale value, or in absolute value.

3.2.1.1 Accuracy rated. - The rated accuracy is a number assigned by the manufacturer to designate the accuracy classification of the voltmeter. The number usually expressed as a percentage of center-scale value, is given as the limit which errors will not exceed when the meter is used under reference conditions.

3.2.1.2 Free indication. - Meters shall be considered as indicating freely if the pointer of the meter under test moves simultaneously with the pointer of the standard meter with which it is being compared, when the electrical energy supplied to both meters is gradually increased from zero to an amount sufficient to produce end-scale deflection of the meter under test.

3.3 Material. - The material for each part shall be as specified herein. When a definite material is not specified, a material shall be used which is adequate for the purpose intended and for meeting the requirement specified herein. Approval or acceptance of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Molded thermosetting plastic material. - Molded thermosetting plastic material conforming to MIL-M-14 shall be used, and shall be of the proper type for its intended use.

3.3.2 Encapsulating material. - Encapsulating material, when used, shall be in accordance with MIL-E-917.

3.4 Design and construction of component parts. -

3.4.1 Indicators. -

3.4.1.1 Three and one-half inch indicators. - 3-1/2-inch indicators shall conform to MIL-M-10304, except with a 250 degree scale length.

3.4.1.2 Four and one-half inch indicators. - 4-1/2-inch indicators shall conform to MIL-M-16034.

3.4.2 Cases. - Voltmeter cases shall be made of molded thermosetting plastic material, or metal and shall be protected against corrosion in accordance with MIL-E-917.

3.4.2.1 Finish. - The portion of the case exposed to view from the front of the panel shall have a gray finish in accordance with formula No. 11, class 2 of MIL-E-15090. No nickel or other bright trimming shall be used. Metal cases shall be rendered resistant to corrosion prior to application of the final finish.

3.4.3 Expanded scale circuit. - Components used in the expanded scale circuit shall be in accordance with MIL-E-917 unless otherwise specified herein.

3.4.4 Scale. -

3.4.4.1 Length. - The scale length shall be 250 degrees, and as specified in 3.19.

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3.4.4.2 Divisions. - The value of each scale division shall be as specified by table I. The angle subtended by a scale division shall be not less than 1 degree.

Table I - Scale divisions

Range	Size of indicator	
	3-1/2 inches	4-1/2 inches
	Volts	Volts
110-120	0.1	0.1
115-125	.1	.1
198-218	.2	.2
210-230	.2	.2
420-460	.5	.5
430-470	.5	.5

3.4.4.3 Mechanical zero point. - The mechanical zero point shall be identified on the scale. The major division mark corresponding to the mechanical zero point shall be extended and identified as "Zero Adjust". The voltage marking at that division should not be disturbed. The marking shall be in red.

3.4.5 Zero adjustment. - Voltmeters shall have a zero-adjuster accessible from the front of the case. It shall provide for a range of adjustment above and below the zero point of not less than 3 percent of the scale length, and preferably shall be capable of rotating through 360 degrees without damage to any part of the instrument, or without becoming inoperative. If the zero-adjuster is not designed for 360 degree rotation, it shall be capable of withstanding a torque of 10 ounce-inches without damage to the zero-adjuster or any part of the instrument. The zero-adjuster shall have sufficient friction to prevent it from shaking loose and changing adjustment when subjected to tests specified herein.

3.4.6 Terminals. - Terminal studs shall be firmly anchored to the base of the meter in such a manner as to prevent turning or movement sufficient to cause damage to internal construction. Each terminal stud shall be furnished with the necessary nuts or screws, and washers (see figures 1 and 2).

3.4.7 Self-contained limitations. - Voltmeters furnished under this specification shall be self-contained.

3.4.8 Mounting. - Voltmeters shall be mounted on a panel drilled in accordance with the applicable drilling plan (see figures 1 and 2). Meters shall be mounted by means of mounting bolts or studs. The necessary mounting bolts (screws), nuts, and lock-washers shall be supplied with each meter.

3.5 Calibration. -

3.5.1 Position. - Meters shall be calibrated in the position in which they are normally used and shall meet the specified accuracy requirements in such position at the reference temperature.

3.5.2 Mounting panel. - Voltmeters shall meet the specified accuracy requirements when mounted on a magnetic steel panel 1/8 inch thick.

3.5.3 Frequency. - Voltmeters shall be calibrated at a nominal frequency of 400 cps.

3.6 Initial accuracy. - The initial accuracy shall be within the limits specified (see 3.22 and 4.3.3).

3.7 Harmonic content. - The change in indication caused by an abnormal waveform described in 4.3.15 shall not exceed the value specified (see table II).

3.8 Frequency range. - Voltmeters shall operate over range specified, and the change in indication from the reading at the nominal frequency shall not exceed the value specified (see table II and 4.3.6).

3.9 On-off cycling. - Meters shall withstand on-off cycling as specified in 4.3.7 without change in indication exceeding that specified in table II.

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3.10 Overvoltage. - Meters shall withstand application of 135 percent rated voltage (center-scale value) for 15 minutes without damage or any permanent change in indication.

3.11 External field influence. - Voltmeters shall have sufficient magnetic shielding so that the effect on an external magnetic field of 5 oersteds will not cause a change in indication greater than the value specified (see table II and 4.3.9).

3.12 Temperature range. - Voltmeters shall indicate freely (see 3.2.1.2) during and after the test specified in 4.3.10. The permanent change in indication shall not exceed the values specified (see table II).

3.13 Inclination. - Voltmeters shall indicate freely during the test specified in 4.3.5. The change in indication shall not exceed the value specified (see table II).

3.14 Vibration. - When voltmeters are tested in accordance with 4.3.13, no screws, bearings, pivots or other parts shall be so loosened or damaged as to cause a change in indication greater than the value specified. (see table II).

3.15 Shock (high-impact). - When voltmeters are tested in accordance with 4.3.14 no screws, bearings, pivots, windows, or other parts shall become loosened, cracked, or unduly damaged. The permanent change in indication based on calibration made after resetting the pointer to zero shall not exceed the value specified (see table II).

3.16 Dielectric stress. - When tested in accordance with 4.3.11 voltmeters shall withstand a voltage of 2600 volts ac without damage of flashover.

3.17 Insulation resistance. - The insulation resistance of voltmeters shall be not less than 20 megohms (see 4.3.12).

3.18 Marking. - Meters shall be permanently and legibly marked with the applicable information specified in 3.18.1.

3.18.1 Dials. - Dial shall be sharply defined and visible from the front of the case. These markings shall not however, be so conspicuous as to distract attention from the scale markings. The scale markings and numerals and items (a), (b), (c) and (d) specified hereinafter shall be marked on the dial; items (a) through (h) shall be marked on the dial, on the front of the meter or on an attached identification plate conforming to MIL-STD-130:

(a) Designation of the quantity measured (volts) (see 6.5).
(b) Frequency (400 c.p.s.).
(c) Type designation (see 1.2.1).
(d) MIL-V-23151.
(e) Manufacturer's name or trade-mark (or symbol), or both.
(f) Manufacturer's model number.
(g) Federal stock number when applicable (see 6.2).
(h) Manufacturer's serial number.

3.19 Scale length. - Scale lengths of instruments shall be as follows:

(a) 3-1/2 inch round 250 degree scale indicator, 5 inches, minimum.
(b) 4-1/2 inch rectangular 250 degree scale indicator, 6.5 inches, minimum.

3.20 Burden. - Burden of meters shall not exceed 10-volt amperes.

3.21 Performance. - Basic initial accuracy of meters shall not exceed 0.1 percent of center-scale value within the center-scale spans of table II for the respective center-scale values and shall not exceed 0.2 percent at all other scale values within the ranges specified in 1.2.1.4, as applicable. Maximum permissible errors are listed in table II in absolute values (volts, plus or minus) for the respective reference conditions.

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Table II - Permissible error, plus or minus, volts

Condition	Center scale values											
	115		120		208		220		440		450	
Scale spans	113-117	110-113 117-120	118-122	115-118 122-125	204-212	198-204 212-218	216-224	210-216 224-230	432-448	420-432 448-460	442-458	430-442 458-470
Initial accuracy	0.115	0.23	0.12	0.24	0.208	0.416	0.22	0.44	0.44	0.88	0.45	0.90
Frequency range	0.0575	0.115	0.06	0.12	0.104	0.208	0.11	0.22	0.22	0.44	0.225	0.45
On-off cycling	0.115	0.23	0.12	0.24	0.208	0.416	0.22	0.44	0.44	0.88	0.45	0.90
Temperature range	0.115	0.23	0.12	0.24	0.208	0.416	0.22	0.44	0.44	0.88	0.45	0.90
Harmonic content	0.1725	0.345	0.18	0.36	0.312	0.624	0.33	0.66	0.66	1.32	0.675	1.35
External field influence	0.0575	0.115	0.06	0.12	0.104	0.208	0.11	0.22	0.22	0.44	0.225	0.45
Inclination	0.0575	0.115	0.06	0.12	0.104	0.208	0.11	0.22	0.22	0.44	0.225	0.45
Vibration	0.115	0.23	0.12	0.24	0.208	0.416	0.22	0.44	0.44	0.88	0.45	0.90
Shock	0.6	1.2	0.6	1.2	1.2	2.4	1.2	2.4	2.4	4.8	2.4	4.8

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3.22 Workmanship. - Meters shall be manufactured in a careful manner in accordance with good design and sound practice. The interiors of meters shall be free from metal fillings, grease or oil, foreign material, dust or other loose particles which may interfere with normal operation of the voltmeter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspection. - The methods of sampling, examination and tests conducted on voltmeters shall be classified as follows:

(a) Qualification (see 4.3).
(b) Quality conformance (see 4.5).

4.2 Test conditions. -

4.2.1 Atmospheric conditions for qualification tests. - Unless otherwise specified herein qualification tests shall be made at ambient temperature of 23° to 27° C., at existing barometric pressure, and with relative humidity of 40 to 60 percent.

4.2.2 Atmospheric conditions for quality conformance tests. - Unless otherwise specified herein, all quality conformance tests shall be made at ambient temperature of 20° to 30° C with a relative humidity of less than 80 percent.

4.2.3 Position. - Unless otherwise specified herein, tests performed on voltmeters, shall be performed with the meter in the position in which it is normally used.

4.2.4 Tapping. - When taking voltmeter readings, either the meter or its support shall be tapped lightly, except when ascertaining the effect of vibration.

4.2.5 Parallax. - Care should be excercised to avoid the effect of parallax when taking meter readings.

4.2.6 Resetting to zero. - Unless otherwise specified herein, meters may be set to zero with use of zero-adjuster at the completion of each test.

4.2.7 Thermal stability. - For all tests, care shall be taken to obtain thermal stability at the specified temperature prior to reading the meter. Thermal stability will be considered attained when the temperature of the meter does not vary by more than one degree per hour.

4.2.8 Number of scale readings. - Unless otherwise specified herein, readings shall be taken at each major division within the center-scale spans given in table II, for example, 113-117 volts. In addition, two additional readings shall be taken at the extreme ends of the scale.

4.2.9 Test equipment. - Standard laboratory instruments shall be used to check the accuracy of indications of meters submitted for test. Approval of the laboratory instruments by Naval Ship Engineering Center (NAVSEC) shall be required prior to their use in checking the accuracy of the meters covered by this specification.

4.3 Qualification tests[1/]. - Qualification tests shall be conducted at a laboratory satisfactory to NAVSEC. Qualification tests shall consist of the tests specified in 4.3.1.

4.3.1 Samples. - Samples submitted for qualification may be either representative of the manufacturer's normal production or a custom made sample. Two meters of each size (3-1/2 or 4-1/2 inches) for which qualification is desired shall be subjected to the examination and tests specified in table III. Examination

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1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

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and tests shall be conducted in the order listed. Meters may be of any voltage rating.

Table III - Qualification examination and tests

Examination and tests	Requirement paragraph	Examination and test paragraph
Visual and mechanical examination (without opening case)	----	4.4
Zero-adjustment	3.4.5	4.3.2
Initial accuracy	3.6	4.3.3
Burden	3.20	4.3.4
Frequency range	3.8	4.3.6
On-off cycling	3.9	4.3.7
Overvoltage	3.10	4.3.8
Inclination	3.13	4.3.5
External field influence	3.11	4.3.9
Harmonic content	3.7	4.3.15
Temperature range	3.12	4.3.10
Dielectric stress	3.16	4.3.11
Insulation resistance	3.17	4.3.12
Vibration	3.14	4.3.13
Shock (high-impact)	3.15	4.3.14
Visual and mechanical examination (external and internal)	----	4.4

4.3.2 Zero adjustment. - The range of zero adjustment above and below the center of the zero mark shall be determined. Record shall be made as to whether or not the zero adjuster may be rotated through 360 degrees without damage to the meter (see 3.4.5).

4.3.3 Initial accuracy. - Voltmeters shall be conditioned in the test circuit until thermal stability is attained while energized to center scale deflection. Readings shall be taken and compared to the readings of the laboratory standard instrument. The test shall be made with the meter at a temperature 45° plus or minus 2°C.

4.3.4 Burden. - The burden shall be measured at center scale deflection and recorded in terms of current voltage, volt amperes, or impedance as applicable.

4.3.5 Inclination. - Voltmeters shall be mounted in the inclined positions noted below and a check for accuracy made at each position.

(a) 60 degrees forward of normal
(b) 60 degrees backward of normal
(c) 60 degrees to right of normal
(d) 60 degrees to left of normal

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4.3.6 Frequency range. - Voltmeters shall be checked for accuracy at frequencies of 395 and 405 cps. The number of readings taken at each frequency shall be as specified in 4.2.8.

4.3.7 On-off cycling. - Meters shall be subjected to a test consisting of 4000 cycles of application and removal of center-scale voltage (plus or minus one volt). The repetition rate of these cycles shall be sufficiently low so that the meter indicator momentarily comes to rest upon both application and removal of voltage. However repetition rates may be used in those cases where this test may cause excessive heating of the meter movement or scale expansion circuit. The change in indication as a result of this test shall be determined.

4.3.8 Overvoltage. - Meters shall be subjected to 135 percent center-scale voltage for 15 minutes. Permanent change in indication as a result of this test shall be determined.

4.3.9 External field influence. - The voltmeter shall be placed at the center of a four turn coil 50 centimeters in diameter, carrying a current of 50 amperes of the same kind and frequency as that which actuates the meter mechanism and which produces a field of approximately 5 oersteds at the center of the coil. Readings shall be taken with the meter at various positions in the center of the field (see 3.11).

NOTE: It is the intent to specify a test coil which will produce a uniform field of approximately 5 oersteds through the case of the meter. Therefore coils of other sizes are equally suitable if an equivalent magnetic field is produced.

4.3.10 Temperature range. - Voltmeters shall be maintained at each extreme temperature (0° to 65°C) for a sufficient time to attain thermal stability (approximately 2 hours). At the end of this period, readings shall be taken at each temperature. The effect of temperature shall not exceed the values specified (see 3.12).

4.3.11 Dielectric stress. - When performing this test, meters with nonmetallic cases shall be mounted on a metallic panel. Voltmeters shall be subjected to a voltage of 2,600 volts rms applied between all external live parts of the meter and:

(a) The metallic platform or panel on which the meter is placed or mounted or the meter case, if the case is metallic.
(b) The zero adjuster.
(c) All exposed metal parts not a live part of the meter.

The voltage shall be nominal 60 cps in frequency and shall approximate a true sine wave in form. The voltage shall be applied continuously for a period of 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer, or by means of an auxiliary ratio transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually, and upon completion of the 60 second test period, shall be gradually reduced. The test shall be made using high potential test rods having not less than 1/8 inch radius tips. Any damage or flashover shall be noted.

4.3.12 Insulation resistance. - Voltmeters shall be placed on a metallic platform or mounted on a metallic panel as specified in 4.3.11. Using a potential of 500 volts direct current the insulation resistance shall be determined between all external live parts and the metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic.

4.3.13 Vibration. - Voltmeters shall be subjected to vibration in accordance with type I of MIL-STD-167. The change in indication shall be determined. Reset to zero will be permitted prior to the determination of the effect of vibration.

4.3.14 Shock (high-impact). - Voltmeters shall be subjected to high-impact shock tests in accordance with type C of MIL-S-901 for light weight equipment when mounted on fixture 6D-1 as shown on figure 7-1 of MIL-S-901. Both test samples shall be subjected to the test. During the test, one voltmeter shall be energized to the center scale value. The pointer may be reset to zero after each blow. At the completion of the shock test, the meters shall be checked for calibration errors, then disassembled and examined for damage (see 3.15).

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4.3.15 Harmonic content. - The following rated waveforms shall be applied to the meter. Accuracy shall be checked.

(a) Fundamental plus 2-1/2 percent each second and third harmonics.
(b) Fundamental plus 2-1/2 percent each second and fourth harmonics.
(c) Fundamental plus 2-1/2 percent each third and fifth harmonics.

4.4 Visual and mechanical examination. - Voltmeters shall be visually and mechanically examined to determine conformance with the examination specified in table IV.

Table IV - Visual and mechanical examination

Requirement	Paragraph
Material	3.3, 3.3.1 and 3.3.2
Design and construction	3.4
Cases	3.4.2 and 3.4.2.1
Mounting	3.4.8
Marking	3.18
Workmanship	3.22

4.5 Quality conformance inspection. - All voltmeters shall be subjected to the examination and tests specified in table V. Voltmeters which do not meet the specified requirements shall be rejected. Reworking of a rejected voltmeter shall be subject to the approval of NAVSEC.

Table V - Quality conformance inspection

Examination and test	Requirement paragraph	Examination and test paragraph
Visual and mechanical examination (without opening case)	---	4.4
Initial accuracy	3.6	4.3.3
Temperature range	3.12	4.3.10
Frequency range	3.8	4.3.6
Dielectric stress	3.16	4.3.11
Insulation resistance	3.17	4.3.12

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

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5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging and packing. - The equipment shall be preserved and packaged by levels A or C and packed by level A or B in accordance with MIL-E-17555.

5.2.2 Marking. - In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use. - Voltmeters covered by this specification are intended for use in type III regulated 400 cycle power systems.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type designation (see 1.2.1).
(c) Federal stock number (see 3.18.1).
(d) Preservation, packaging, packing and marking, if other than specified in 5.1 (see 5.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Precautions. - Attention of equipment designers and manufacturers is invited to the fact that sufficient space should be provided in equipment for mounting standard meters of the maximum depth specified (see figures 1 and 2) even though shallower meter may be in the initial design.

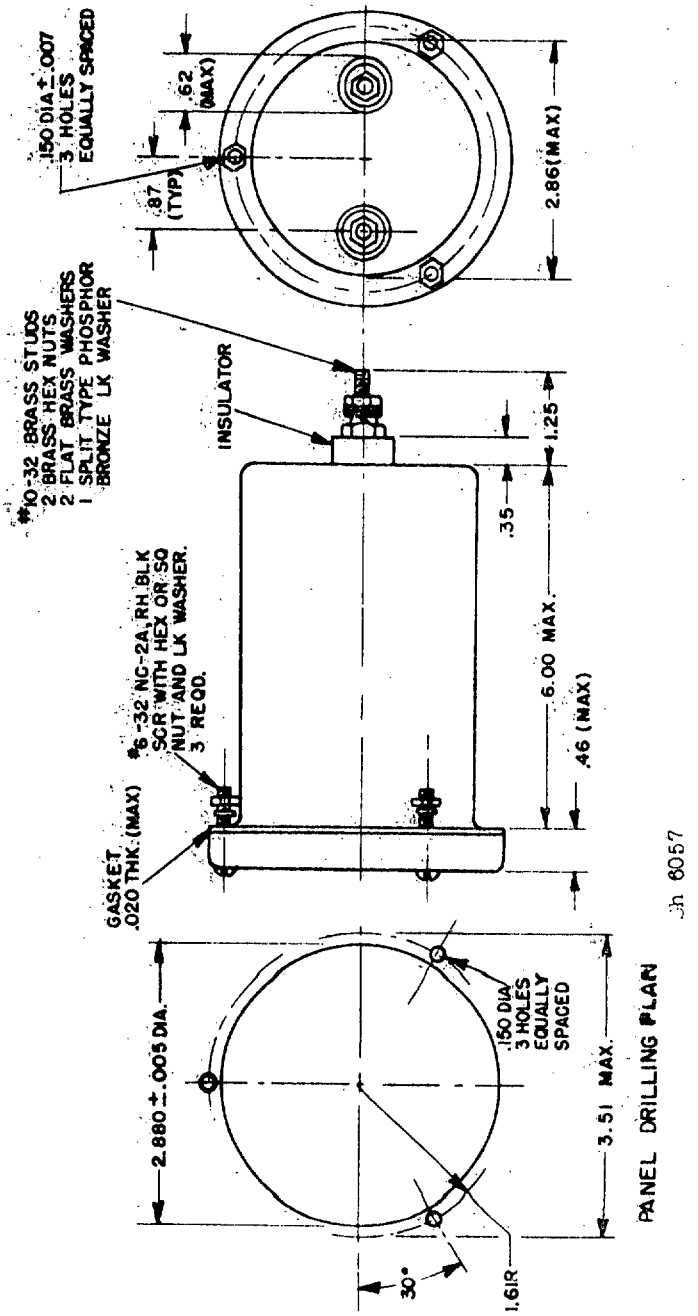
6.4.1 Red mark on dial. - When specified in the contract or order, the manufacturer may place red marks on meter dials at designated positions, there will be no change in the type designation to distinguish such meters from those having normal markings; consequently, a meter supplied for a certain application may have a red mark at an undesired position. If this occurs, it is recommended that the user remove or cover the red mark by any convenient method, and make other red marks, if desired.

6.5 Meters produced against contracts or orders placed with the meter manufacturer on or before 31 December 1967 shall be marked "Cycles (Hertz)". Meters produced against contracts or orders placed on or after 1 January 1968 shall be marked "Hertz".

6.6 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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Figure 1. - Size and mounting dimensions, MR 39 series.

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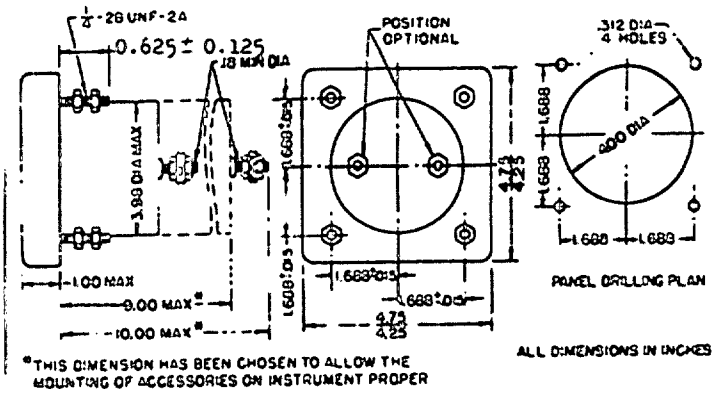


Figure 2 - Size and mounting dimensions, MR 49 series.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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